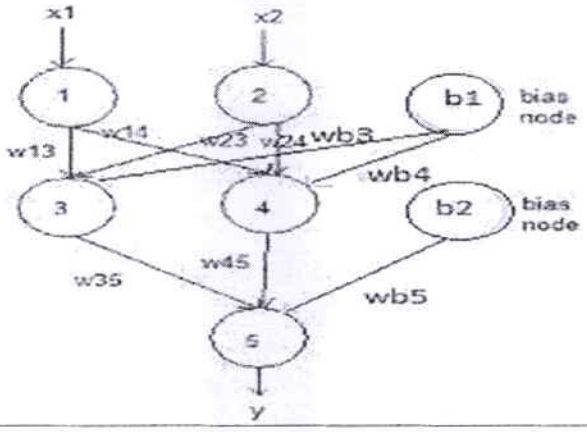
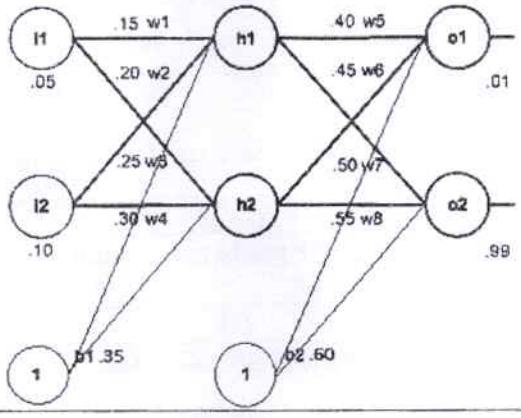
Roll No……..

Mid Semester Examination, March. 2025

Name of the Program: B. Tech. ***Semester:*** *IV*
Name of the Course: Deep Learning ***Course Code:*** TCS 464

Time: 1-1/2 Hour ***Maximum Marks***: 50

Note:
(i) Answer **all the questions** by choosing ***any one of the sub questions***.
(ii) Each question carries 10 marks.

Q1		10 Marks
(a)	In a multi-perceptron layer (MLP), compute forward and backward propagation for epoch 1 (assume learning rate as 0.3). 	CO1
	OR	
(b)	For the given scenario calculate the total error after forward pass 	CO1
Q2		10 Marks
(a)	What is McCulloch-Pitts Neuron? Solve XOR gate using MP neuron.	CO2

OR

(b) We have a classification problem with four classes of input vector. The four classes are — CO2

$$\text{class 1: } \left\{ \mathbf{p}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{p}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}, \text{ class 2: } \left\{ \mathbf{p}_3 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \mathbf{p}_4 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\},$$

$$\text{class 3: } \left\{ \mathbf{p}_5 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \mathbf{p}_6 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}, \text{ class 4: } \left\{ \mathbf{p}_7 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \mathbf{p}_8 = \begin{bmatrix} -2 \\ -2 \end{bmatrix} \right\}$$

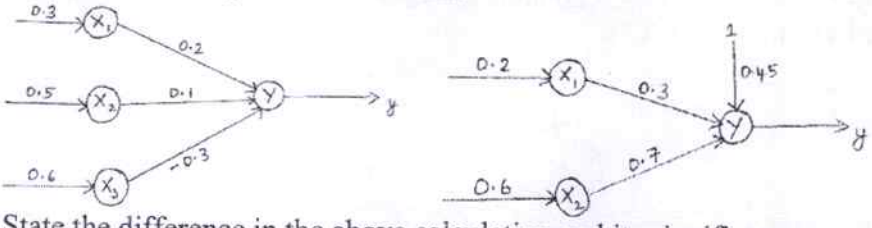
Design a perceptron network to solve this problem.

Q3 — 10 Marks

(a) What is the need on modification in perceptron? Explain the mathematical significance of sigmoid function in neural network. — CO2

OR

(b) Calculate the net input for the networks. — CO2



State the difference in the above calculation and its significance.

Q4 — 10 Marks

(a) Explain how loss functions work? List at least three loss functions (with mathematical notations) which can be applicable to the classification problems. — CO3

OR

(b) Apply hinge loss and cross-entropy loss for the given matrix: — CO3

$$\underbrace{\begin{bmatrix} 0.01 & -0.05 & 0.1 & 0.05 \\ 0.7 & 0.2 & 0.05 & 0.16 \\ 0.0 & -0.45 & -0.2 & 0.03 \end{bmatrix}}_{W} \underbrace{\begin{bmatrix} -15 \\ 22 \\ -44 \\ 56 \end{bmatrix}}_{x_i} + \underbrace{\begin{bmatrix} 0.0 \\ 0.2 \\ -0.3 \end{bmatrix}}_{b}$$

Q5 — 10 Marks

(a) What is convolution in CNN? Write the mathematical notations, what is the use of max pooling. Explain with example. — CO3

OR

(b) What are challenges of deep learning algorithms? Explain with terms learning rate and momentum. — CO3